

CHILUKA MANIRATHNAM

Cybersecurity Engineer | Offensive Security Researcher | Application Security Enthusiast

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LinkedIn | Hack The Box | GitHub |

PROFESSIONAL SUMMARY

Cybersecurity student specializing in Application Security, Network Penetration Testing, Offensive Security Research, and Security Automation. Experienced in web application security testing, API security assessments, malware analysis, SDR-based hardware security research, and vulnerability discovery. Certified APISEC Practitioner and Junior Web Application Penetration Tester with hands-on expertise in Python, Linux, Burp Suite Professional, Nmap, Wireshark, and offensive security methodologies.

TECHNICAL SKILLS

Security Domains: Application Security, API Security, Web Security, Network Security, Wireless Security, Vulnerability Assessment, Malware Analysis, Digital Forensics, Incident Response

Security Tools: Burp Suite Professional, Nmap, Wireshark, Metasploit Framework, Gobuster, Hydra, John the Ripper, tcpdump, HackRF One, Flipper Zero

Programming: Python, C, Java, Bash Scripting

Platforms: Kali Linux, Oracle VirtualBox, Hack The Box, TryHackMe

Core Concepts: OWASP Top 10, API Security Testing, Network Enumeration, Privilege Escalation, Security Automation, Threat Detection, Packet Analysis

CERTIFICATIONS

- **Certified Junior WebApp Penetration Tester (JWPT)** May 2026 Hackers Daddy
- **APISEC Certified Practitioner (ACP)** Apr 2026 APISEC University (Credly Verified)
- **Machine Learning for Cyber Security** Feb 2025 C-DAC Hyderabad
- **Enterprise Routing and Security Stack** Jan 2026 Cisco Networking Academy
- **TryHackMe Advent of Cyber Certification** Dec 2025 Credential ID: THM-OQXK6YS6AA

TECHNICAL PROJECTS AND SECURITY RESEARCH

Rolling Code Interception and Frequency Replay Framework 2026

- Designed and implemented a Software Defined Radio (SDR) framework to evaluate vulnerabilities in sub-GHz rolling-code authentication systems.
- Simulated RollJam-style replay attacks using HackRF One and Flipper Zero to demonstrate signal interception and delayed transmission attacks.
- Documented mitigation techniques including anti-replay validation, rolling counter synchronization, and secure token management mechanisms.

Automated PHP Reverse Shell Static Code Analyzer 2025

- Developed a Python-based static analysis engine capable of detecting malicious PHP web shells and obfuscated backdoor scripts.
- Implemented heuristic and regex-based detection techniques targeting dangerous functions such as `exec()`, `system()`, `passthru()`, and `shell_exec()`.
- Automated alert generation and logging workflows to improve visibility into potentially compromised web directories.

Practical Cybersecurity Labs (Hack The Box and TryHackMe) 2025 – Present

- Completed 24+ cybersecurity challenge environments covering web exploitation, OSINT, Linux privilege escalation, and network forensics.
- Performed structured host enumeration, vulnerability discovery, credential attacks, and post-exploitation activities within simulated enterprise environments.
- Utilized Burp Suite Professional, Gobuster, Hydra, Nmap, and Wireshark during offensive security assessments.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science Engineering (Cybersecurity)

CMR College of Engineering and Technology, Hyderabad

2024 – 2027

CGPA: 7.5 / 10

Diploma in Computer Science Engineering

Kshatriya College of Engineering

2021 – 2024

GPA: 8.29 / 10

Secondary School Certificate (SSC)

TS Social Welfare Residential School

2021

GPA: 10.0 / 10

PROFESSIONAL DEVELOPMENT

- Hack The Box Certified Penetration Testing Specialist (CPTS) Path – Currently in Progress
- Studying Active Directory Enumeration, Privilege Escalation, Web Application Exploitation, Pivoting, and Penetration Testing Methodologies
- Conducted vulnerability assessments, web application testing, network reconnaissance, and privilege escalation exercises.
- Practiced OWASP Top 10 testing methodologies and API security validation techniques.
- Experienced with Linux administration, packet analysis, and offensive security tooling.